Project Proposal - Deep Learning and Multimodal Systems

PharmaTrade-MM: Multimodal RL Agent for Pharma Stock Trading Using FDA Event Signals

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1. Goal of the Project:

Most RL trading agents only use price data, or at best add generic news sentiment. In pharma, the biggest single-day price swings come from FDA decisions (drug approvals, rejections, advisory committee outcomes) which are scheduled months in advance and can move a stock 30–50% in one session. We build a domain-specialized multimodal RL agent that trades eight major pharma stocks (PFE, JNJ, MRK, ABBV, BMY, AMGN, GILD, BIIB) using three input streams: price and technical indicators, pharma news sentiment via FinBERT, and a structured FDA event calendar encoding regulatory decisions, drug pipeline stage, therapeutic area, and historical price reactions. We extend Nawathe et al. (2024), which uses SEC filings as its third modality, arguing that backward-looking quarterly filings are far weaker than forward-looking FDA catalysts for pharma stocks. We also add a CVaR-augmented reward with event-aware position sizing to handle tail-risk around FDA events.

2. Benefits:

- **Domain gap:** Almost no prior RL trading work specializes in pharma, where binary FDA events dominate price behavior.
- **Novel modality:** No published RL trading paper encodes drug approval calendars as a structured modality. Our team's pharma background makes this credible and well-grounded.
- **Risk-aware design:** CVaR reward + event-aware position sizing directly addresses the rare-but-catastrophic loss problem at FDA decisions.
- **Clean ablation:** Each modality addition is independently testable, producing directly interpretable results on what drives performance gains.

3. Data Set Details:

Modality	Source	Coverage / Access
Price + Technicals (OHLCV, RSI, MACD, BB)	Yahoo Finance (yfinance)	2018–2023 daily / Free
Pharma News Sentiment	SEC EDGAR 8-K filings + Reuters pharma headlines (Kaggle)	Per-ticker daily / Free
FDA Event Calendar (novel modality)	BioPharma Catalyst (Kaggle) + FDA.gov; enriched with pipeline stage, therapeutic area, historical reaction	PDUFA/AdCom/CRL 2018–2023 / Free
Benchmark Returns	XPH Pharma ETF via yfinance	2022–2023 test window / Free

4. Project Method:

The system has three components: a multimodal state encoder, a PPO actor-critic network, and a custom OpenAI Gym trading environment. The encoder uses a 2-layer LSTM (hidden 128) for price features, a 1-layer LSTM (hidden 64) for daily FinBERT sentiment, and a small MLP for the FDA event vector (days-to-event, event type one-hot, therapeutic area, historical reaction → 64-dim). These are fused via cross-

attention (price attends over sentiment) then concatenated with the FDA embedding and projected to a 256-dim state. The PPO actor outputs Buy/Sell/Hold probabilities; the critic outputs a value estimate. The reward is an incremental Sharpe ratio minus a CVaR tail-risk penalty, drawdown penalty, and 0.1% transaction cost per trade. In the five days before any scheduled FDA decision, a position-scaling factor additionally reduces exposure. We run five ablation variants: (1) Price only, (2) Price + Sentiment, (3) Price + Sentiment + SEC Filings (replicating Nawathe et al.), (4) Price + Sentiment + Basic FDA, and (5) Price + Sentiment + Rich FDA + CVaR (our full model). All are backtested on held-out 2022–2023 data.

5. Milestones

- Data collection: price download (yfinance), FinBERT sentiment pipeline, FDA event calendar curation (pharma teammates)
- Build Gym trading environment, implement CVaR reward function, sanity-check with random agent
- Build multimodal encoder + PPO actor-critic; first training run on price-only baseline (Colab Pro A100)
- Train all five ablation variants; backtest on 2022–2023 held-out data; FDA event proximity interpretability analysis
- Compute all evaluation metrics, generate plots, draft report sections
- Report finalization, presentation prep, submission

6. Evaluation Criteria:

We backtest on the held-out 2022–2023 window and compare against: buy-and-hold, equal-weight monthly rebalance, 20-day momentum strategy, XPH pharma ETF, and each ablation variant. Primary metrics: Annualized Sharpe Ratio (risk-adjusted return), Maximum Drawdown (worst peak-to-trough loss), Sortino Ratio (downside-only volatility penalty), Cumulative Return vs. benchmark, Alpha vs. XPH ETF, and Win Rate. We also include an interpretability analysis plotting agent position size versus days-to-FDA-event; if the agent learned to reduce exposure pre-event, that validates the FDA modality is being meaningfully used.

7. Deliverables:

- **Code:** Full training + evaluation codebase on public GitHub (data pipeline, environment, model, training, evaluation modules) with README and reproducibility guide.
- **Data:** Processed features (price, sentiment, FDA calendar) published with repo. All raw sources are free and publicly accessible (Section 3).
- **Report:** Full course report covering related work, methodology, experiments, results, and analysis in course-specified format.
- **Presentation:** Slides with ablation results, portfolio performance curves, and interpretability plot. Live demo of trained agent inference where compute permits.

8. Dependencies and Requirements:

- **Compute:** TACC/Google Colab Pro (A100 GPU). Each ablation variant should train in ~20–40 min on A100; full ablation feasible in one day.
- **Libraries:** PyTorch, Stable-Baselines3 (PPO), Hugging Face Transformers (FinBERT), yfinance, QuantStats, OpenAI Gym.
- **Data access:** All sources are publicly available with no restricted-access agreements required.

References

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[3] Liu, X. et al. (2024). SAPPO: Sentiment-Augmented Proximal Policy Optimization for Algorithmic Trading. arXiv preprint.

[4] Fatemi, S. et al. (2025). Cross-Modal Temporal Fusion for Financial Market Forecasting. arXiv:2504.13522.

[5] Loughran, T. & McDonald, B. (2011). When Is a Liability Not a Liability? Journal of Finance, 66(1), 35–65.

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